Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/24/2010
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Investigations of the Plant Cytoskeleton in Microgravity With Gene Profiling and Cytochemistry		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences BRIC NNH09ZTT004N
Start Date:	02/08/2010	**End Date:**	02/07/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University)		
Grant/Contract No.:	NNX10AF44G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goal of this research is to study the effects of microgravity on the structure and organization of actin cytoskeleton in plants. This proposed project builds on our previous ground-based and spaceflight research using the model plant Arabidopsis. Thus, the specific aims of this proposed flight research using the BRIC-LED system are: (1) to investigate plastid position in statocytes (=gravity-perceiving cells) in microgravity; (2) to determine the effect of microgravity on the actin cytoskeletal organization in gravity-perceiving cells; (3) to study microgravity effects on actin cytoskeleton-related gene expression in plant cells. This project also will allow us to directly correlate results from cytological investigations and gene profiling in order to understand the nature of the actin cytoskeleton in mechanisms of gravity perception. This proposed research focuses on the effects of gravity on basic cellular mechanisms and processes in plants, so it is relevant to the emphasis on using Arabidopsis as a model system in developmental biology highlighted in the current NRA. The PI's group has extensive experience with the use of Arabidopsis in experiments on the Space Shuttle and the ISS. Preliminary data on the growth of seedlings in the BRIC-LED system and from the gene profiling studies performed during our recent experiments on the ISS are presented to show the feasibility of the proposed project. Improved knowledge of the basic mechanistic processes that will be the focus of this research is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposed project on the analysis of the role of the actin cytoskeleton in mechanisms of gravity perception builds on our previous ground- and flight-based research. We plan to study the effects of gravity on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology as highlighted in the current NASA Research Announcement (NRA). Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our space flight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture.
Task Progress:	The NASA Biological Research in Canisters (BRIC) -16 project offered a unique opportunity to conduct a spaceflight experiment with minimal preparation time. Procedures were adapted from previous spaceflight studies conducted by our lab on the Space Shuttle and the ISS. Seedlings of Arabidopsis thaliana were grown from seed on the shuttle Discovery mission STS-131 in the BRIC- Petri Dish Fixation Unit (PDFU) flight hardware. After fourteen days of growth, samples were then stabilized by either an aldehyde-based fixative or RNAlater. We compared flight samples with a ground sample control, which was conducted in the flight hardware at 1g. Through additional studies, samples will be analyzed using molecular genetic techniques and three methods of microscopy including light (LM), transmission electron (TEM), and confocal. LM will be used to understand any overall anatomical differences and the placement of the statoliths. TEM will elucidate the interaction between statoliths and the sub-cellular structures such as cytoskeleton and organelles. Confocal microscopy will show us the structural differences in the development of the actin cytoskeleton between ground and flight. The overall goal of this study is to correlate results from gene profiling with the cytological investigations conducted through these three microscopy techniques to promote a more complete understanding of the effects of gravity on basic cellular mechanisms and processes in plants.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Johnson C, Kiss JZ, Edelmann RE, Millar KDI, Correll MJ. "The effects of microgravity on the actin cytoskeleton of Arabidopsis thaliana. " Graduate Student Poster Session. 26th annual meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 5-7, 2010. Program and abstracts, 26th annual meeting of the American Society for Gravitational and Space Biology, 2010. p. 33. , Nov-2010
Abstracts for Journals and Proceedings	Kiss JZ. "The use of microgravity to study tropistic responses in plants." 26th annual meeting of the American Society for Gravitational and Space Biology, Washington, DC, Nov. 4-7, 2010. Program and abstracts, 26th annual meeting of the American Society for Gravitational and Space Biology, 2010. p. 17. , Nov-2010
Awards	Johnson C. "Third place: Graduate Student Forum. 'The effects of microgravity on the actin cytoskeleton of Arabidopsis thaliana.' American Society for Gravitational and Space Biology, Washington, DC, November 2010." Nov-2010